

Fundamental Data Types in Python, Looping and Decision-making constructs, functions, classes, file handling, database access, output formatting, classes, modules, statistics module, numpy, introduction to popular machine learning libraries Tensor Flow & Keras.

**Unit-3.0 Unsupervised Learning** **7 hrs.**

Clustering, K-means, GMM & EM Algorithm. Eigen values and Eigen vectors, PCA – unsupervised dimensionality reduction technique.

**Unit-4.0 Supervised Learning** **7 hrs.**

Classification: KNN, Bayes, Decision Tree. SVM: soft and hard margin, kernel trick. LDA – supervised dimensionality reduction technique.

**Unit-5.0 Ensemble and Reinforcement Learning** **7 hrs.**

Bagging, Random Forest and Boosting. Q-learning and SARSA algorithms.

**Unit-6.0 Neural Networks** **7 hrs.**

Introduction, perceptron model, learning rules and activation functions, multi-layer feed forward, back-propagation, introduction to feed-back networks.

**Text/Reference Books**

1. Machine Learning. Tom Mitchell, McGraw-Hill.
2. Machine learning: an algorithmic perspective. Marsland, Stephen. Chapman and Hall/CRC, 2011.
3. Introduction to artificial neural systems. Zurada, Jacek M. Vol. 8. St. Paul: West publishing company, 1992.
4. A Tutorial on Support Vector Machines for Pattern Recognition. Christopher Burges, Data Mining and Knowledge Discovery, 1998

**Course Code- 119306****Materials, Testing & Evaluation****3 0 0 3****Unit- 1.0** **7 hrs**

Introduction to Engineering Materials covering, Cements, M-Sand, Concrete (plain, reinforced and steel fibre/ glass fibre-reinforced, light-weight concrete, High Performance Concrete, Polymer Concrete) Ceramics, and Refractories, Bitumen and asphaltic materials, Timbers, Glass and Plastics, Structural Steel and other Metals.

**Unit- 2.0** **6 hrs**

Paints and Varnishes, Acoustical material and geo-textiles, rubber and asbestos, laminates and adhesives, Graphene, Carbon composites and other engineering